EFY TESTED Ashwini K. Sinha

IoT On The Eyes: To Control Home Appliances

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Have you ever thought of controlling electric/electronic home appliances by simply looking at them? Could happen in movies but seems impossible in real life.

Well, no more. Here is perhaps world's first device that lets you switch on or switch off any electric/electronic appliance with the blink of your eyes. This cool IoT device would also help specially-abled people to independently control such appliances.

Materials — Table 1

Component name	Quantity	Description
Raspberry Pi 4	1	2GB or greater RAM
Rpi Camera	1	Camera module
Raspberry Pi Pico	1	Microcontroller
Bluetooth HC05	1	Bluetooth module
Relay module	1	5V relay module

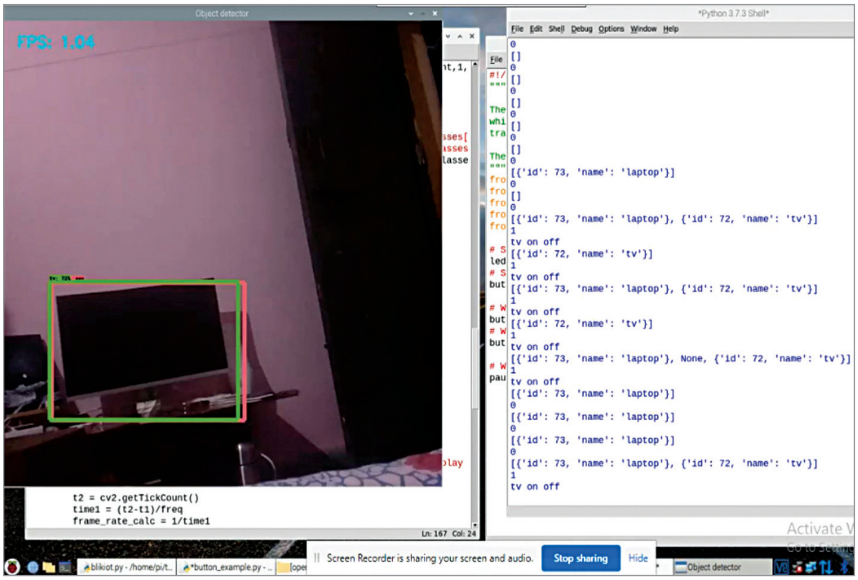

Fig. 2: Testing code

Fig. 1: Author's prototype

Coding

The device needs to recognise the appliances that have to be controlled through commands sent from the eye. Therefore, the following code permits capture of real-time videos for object detection and toggling an appliance on or off.

OpenCV is used to capture video from the camera and TensorFlow (TF) to recognise the home appliances being looked at.

```
import os
import cv2
import numpy as np
from picamera.array import PiRGBArray
from picamera import PiCamera
import tensorflow as tf                    Change
import argparse
import sys
import time
import serial

from signal import pause
import threading
ser = serial.Serial("/dev/rfcomm2",baudrate=9600)

Time=0
a=time.time()
#IM_WIDTH = 880
```

Fig. 3: Code setting serial baud rate and port name

Install Python and the required modules in Raspberry Pi using following commands:

```
sudo pip3 install python-opencv
sudo pip3 install tensorflow
sudo pip3 install keras
sudo pip3 install gpiozero
Sudo pip3 install pyserial
git clone --depth 1 https://github.com/
tensorflow/models.git
```

If you face a problem in installation of TensorFlow and OpenCV, or get an error due to the version of TensorFlow, follow the instructions for installation of TensorFlow and object detection given in TensorFlow website.

Next, clone the object detection library. Then in the test folder, create a file containing the list of appliances and save it as *eyeiot.pbtxt*